Huiqin (Hermine) Ni

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Education

North Carolina State University **Aug 2024 – Present**
Ph.D. in Mechanical Engineering; Advisor: Prof. Arun Kumar Kota *Raleigh, NC, USA*

Aalto University **Sep 2022 – Aug 2023**
Visiting Student, Department of Applied Physics; Advisor: Prof. Olli Ikkala *Helsinki, Finland*

China University of Petroleum **Aug 2021 – Jun 2024**
M.S. in Chemical Engineering and Technology; Advisor: Prof. Quan Xu *Beijing, China*

China University of Petroleum **Aug 2017 – Jun 2021**
B.Eng. in Materials Science and Engineering; Rank: 2/84 *Beijing, China*

Research Interests

Functional surfaces and interfaces; smart and responsive polymers; bioinspired adhesion; electroadhesion; slippery surfaces; soft actuators; additive manufacturing.

Publications

Peer-Reviewed Journal Articles

- A. M. Rather, S. Pillai, A. Nair, M. J. Zarei, Y. J. Kim, **H. Ni**, A. K. Kota. "Liquid crystal elastomer-based reversible metamorphosis for loss-less droplet manipulation." *Materials Horizons* **13**(6), 2946–2957 (2026). doi:10.1039/D5MH01295C.
- Y. Mi, Y. Niu, **H. Ni**, Y. Zhang, L. Wang, Y. Liu, M. A. Ramos, T. S. Hu, Q. Xu. "Gecko-inspired reversible adhesion via quantum dots enabled photo-detachment." *Chemical Engineering Journal* **431**, 134081 (2022).
- Q. Xu, Y. Niu, J. Li, Z. Yang, J. Gao, L. Ding, **H. Ni**, P. Zhu, Y. Liu, Y. Tang, Z. Lv, B. Peng, T. S. Hu, H. Zhou, C. Xu. "Recent progress of quantum dots for energy storage applications." *Carbon Neutrality* **1**, 13 (2022).
- X. Luo, X. Dong, H. Zhao, T. S. Hu, X. Lan, L. Ding, J. Li, **H. Ni**, J. A. Contreras, H. Zeng, Q. Xu. "Near-infrared responsive gecko-inspired flexible arm gripper." *Materials Today Physics* **29**, 100919 (2022).

Manuscript in Preparation

H. Ni, O. Ikkala, Q. Xu. Responsive polymer actuation. Manuscript in preparation.

Book Contributions

Energy Bionics. China Petrochemical Press, 2021. ISBN: 978-7-5114-6097-4.
Liquid Flow Battery and Energy Storage. Sinopec Press, 2022. ISBN: 978-7-5114-6670-9.

Research Experience

Solid Slippery Polymer Coatings **Sep 2026 – Present**
North Carolina State University

- Investigate functional polymer coatings and surface wetting behavior.
- Contribute materials preparation, experimental design, surface characterization, and data analysis.

LCE-Based Slippery Surfaces **Aug 2024 – Present**
North Carolina State University

- Study responsive polymer surfaces and their interfacial behavior as part of Ph.D. research.
- Contributed materials fabrication, experimental design, surface characterization, and data analysis.

Related Surface Engineering Projects

- Contributed to polymer surface-engineering projects, including studies relevant to potential biomedical applications.
- Built experience in materials processing, surface preparation, and evaluation of wetting behavior.

Electroadhesive Polymer and Hydrogel Interfaces **2026 – Present**

- Investigate adhesion at soft polymer and hydrogel interfaces.
- Design comparative experiments and contribute materials characterization and analysis.

Research Experience (continued)

Multi-Mode Shape-Memory Polymer Actuators **Oct 2022 – Jun 2024**
China University of Petroleum / Aalto University
- Investigated responsive shape-memory polymers for soft actuation.
- Conducted materials fabrication, characterization, mechanical testing, and manuscript preparation.

MXene Processing for Printed and Wearable Electronics **Oct 2022 – Jun 2023**
- At Aalto University, compared MXene solvent-exchange conditions for printed and wearable electronics.

Controllable Bioinspired Adhesion for Robotic Grippers **Dec 2020 – Jan 2022**
- Developed gecko-inspired PDMS surfaces; fabricated and characterized samples and drafted the manuscript.

Multi-Field Responsive Hydrogel Actuators **Mar 2020 – May 2022**
- Investigated PNIPAM-based nanocomposite hydrogels for infrared, magnetic, and electric actuation.

Defect-Engineered Bi-Based Photocatalytic Materials **Apr 2019 – Mar 2020**
- Studied oxygen vacancies and heterojunctions in BiOBr photocatalysts for wastewater purification.

Teaching Experience

Graduate Teaching Assistant, NC State University **Fall 2026**
- Teach two undergraduate laboratory sections; guide experiments, data analysis, and technical reporting.
- Grade assignments using departmental criteria; hold office hours for data interpretation and report preparation.

Entrepreneurship & Engineering Projects

Sensor Fused Bus — Robotics and Sensor Integration **2026**
- Develop sensor-integrated robotic platform concepts for commercial applications; conduct customer discovery.
- Translate customer needs into system concepts, engineering trade-off analysis, and bill-of-materials planning.

Industry & Science Communication Experience

Large Language Model Internship Projects **Jan 2024 – Jun 2024**
- Conducted LLM training and evaluation tasks for projects associated with Alibaba and Tencent; tested outputs and provided feedback for product development.

Science Communication Initiative, DeepTech Collaboration **Nov 2021 – Mar 2022**
- Published content on emerging research; maintained an online science community and hosted recurring webinars.

Selected Presentations & Conferences

- International Summer School Materials 4.0, Dresden, Germany (2023): poster and oral presentation, "Multi-modes Inspired from Nature EVA Actuator."
- LIBER Symposium, Helsinki, Finland (2023): biomimetic materials.
- Frontiers of Biological Materials: Principles and Applications, MRS Bulletin / *Nature Materials*, virtual (2023); ICBE, Wuhan, China (2023); ICFE, Hangzhou, China (2023).

Honors & Awards

- Leon Memorial Scholarship, NC State University College of Engineering (Jun 2026).
- Materials 4.0 Summer School Scholarship (2023); Advanced Individual in Science and Technology Innovation (2022); Outstanding Graduate (2021).
- National Student Translation Competition, Second Prize (2019); National College English Contest, Third Prize (2018, 2019); National Talent Cup English Reading Contest, Third Prize (2019).
- Mathematical Contest in Modeling, Successful Participant (2018, 2019); Excellent Volunteer on Campus.

Technical Skills & Training

Fabrication: Polymer synthesis/processing, hydrogel fabrication, nanomaterials, plasma treatment, surface modification, laser processing, coating, 3D printing.

Characterization: Contact/sliding-angle goniometry, FT-IR, AFM, SEM, UV-Vis, XRD, DSC, TGA, DMA, Instron.

Computational Tools: Python, MATLAB, C, Linux/Ubuntu, ns-3, SolidWorks, CAD tools, Abaqus, Origin, LaTeX.

Languages: Chinese (native); English (professional working proficiency); Japanese (basic).

Certification & Training: ACCA Diploma in Accounting and Business; Spring School in Computational Chemistry, Helsinki (2023).